# NATIONAL INSTITUTE OF TECHNOLOGY PATNA

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**End Semester Examination, December 2024**

**Course Code: CH14101 (CSE-I)** **Time: 3.00 Hours**

**Course Title: Engineering Chemistry** **Full Marks: 60**

**Answer all questions**

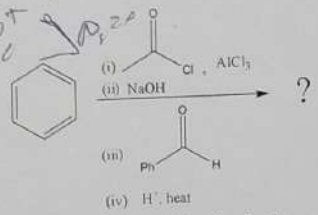
Q1. (a) Write the complete sequence of energy levels in the increasing order of energy in the molecule as per Molecular Orbital Theory. Compare the relative stability and the magnetic behaviour of the following species: $N_2$, $N_2^+$, $N_2^-$, $N_2^{2+}$

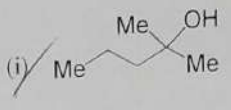
(b) The molecular shape of $[Ni(CO)_4]$ is not the same as that of $[Ni(CN)_4]^{2-}$ Explain why using Valence Bond Theory.

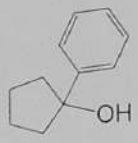
(c) Explain with example Coordination Isomerism and Coordination Position Isomerism.

(d) Calculate the CFSE for $[Fe(CN)_6]^{4-}$ (CO3) 4x5=20

Q2. (a) Write the final product of the reaction with mechanism

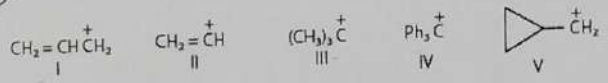



(b) Show how each alcohol can be prepared (i) 

(ii) 


(c) Predict order of reactivity in $S_N1$ and $S_N2$ reaction for four isomeric bromobutanes

(d) Arrange the following carbocations in decreasing order of stability




(CO4) 4x5=20

Q3. (a) Write the Nernst equation for the cell reaction in the Daniell cell. How will the $E_{cell}$ be affected when concentration of $Zn^{2+}$ ions is increased?

(b) Calculate the standard electrode potential of $Ni^{2+}|Ni$ electrode if the cell potential of the cell $Ni|Ni^{2+}(0.01M)||Cu^{2+}(0.1M)|Cu$ is 0.59V. Given that $E^o_{(Cu^{2+}|Cu)} = 0.34V$

(c) For the cells shown below, write cell reactions

(i) $Cd(s)|Cd^{2+}||Ag^+|Ag(s)$

(ii) $Zn(s)|ZnSO_4(aq)||CdSO_4(aq)|Cd(s)$

(iii) $Fe(s)|FeCl_2(aq)||AuCl_3(aq)|Au(s)$

Calculate standard cell potential, if standard reduction potentials are $E^o_{Zn^{2+}/Zn} = -0.76V$; $E^o_{Cd^{2+}/Cd} = -0.40V$; $E^o_{Au^{3+}/Au} = +1.42V$; $E^o_{Fe^{2+}/Fe} = -0.44V$; $E^o_{Ag^+/Ag} = +0.80V$. (CO2) 3x5=15

Q4. Explain (i) Flue Gas Analysis and (ii) Fuel Cell. (CO1) 5

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